

Compute $((3I_2)A)$: Multiply matrix (A) by 3, which scales every element in (A) by 3.

i am having trouble understanding this proof technique where they used variables to denote for all.

I.e., how does the identity matrix I_2 get 3 into all of A ? That logic does not follow, and there are some 0s in I_2

TL;DR

$3I_2$ does **not** copy its zeros into A . The identity matrix acts like the number 1 in matrix multiplication. Multiplying it by 3 turns it into "3 times the identity," which acts like multiplying **every row of A** by 3.

Step-by-step (pure logic, no handwaving)

Let

$$I_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

Then

$$3I_2 = \begin{pmatrix} 3 & 0 \\ 0 & 3 \end{pmatrix}$$

Now multiply:

$$(3I_2)A = \begin{pmatrix} 3 & 0 \\ 0 & 3 \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

Matrix multiplication rule:

$$[(3I_2)A]_{ij} = \sum_{k=1}^2 (3I_2)_{ik} A_{kj}$$

Since I_2 is zero except when $i = k$:

$$(3I_2)_{ik} = 3\delta_{ik}$$

So:

$$[(3I_2)A]_{ij} = \sum_k 3\delta_{ik} A_{kj} = 3A_{ij}$$

Therefore:

$$(3I_2)A = \begin{pmatrix} 3a & 3b \\ 3c & 3d \end{pmatrix} = 3A$$

Why the zeros don't "mess things up"

- The zeros in I_2 **cancel terms**, they don't overwrite entries
- Each row of $3I_2$ has exactly one nonzero entry (a 3)
- That 3 multiplies the corresponding row of A

Think:

- $I_2 A = A$ (identity does nothing)
- $3I_2 A = 3A$ (identity scaled by 3 scales the result)

Why variables are used in proofs

Using symbols like A_{ij} proves the statement for **all matrices**, not one example. That's how linear algebra avoids case-by-case checking.

Sources

- Lay, *Linear Algebra and Its Applications*, Section 2.2 (Matrix Multiplication Properties)
- Strang, *Introduction to Linear Algebra*, Chapter 2
- Wikipedia: "Identity matrix" and "Scalar multiplication"
https://en.wikipedia.org/wiki/Identity_matrix ↗
https://en.wikipedia.org/wiki/Matrix_multiplication ↗

If you want, I can rewrite this proof in even more mechanical index notation only.



i know the definition of for all, what i am asking is the logical jump to say (oh since we have pivot 3 in each column, thus we multiply the pivot by each vector within A)

TL;DR

There is no "pivot-logic" rule being applied. The only valid jump is the **definition of matrix multiplication**. The "pivot 3" language is an *interpretation*, not a reason.

Where the confusion is coming from

This sentence is **not** a logical step:

"since we have pivot 3 in each column, we multiply each vector in A by 3"

That is **intuition after the fact**, not proof.

The proof uses **only the multiplication rule**.

The only legitimate step

Let A be any $2 \times n$ matrix.

$$(3I_2)_{ik} = 3\delta_{ik}$$

Matrix multiplication:

$$[(3I_2)A]_{ij} = \sum_k (3I_2)_{ik} A_{kj}$$

Substitute:

$$= \sum_k 3\delta_{ik} A_{kj}$$

Key logical fact (this is the whole proof):

$$\sum_k \delta_{ik} A_{kj} = A_{ij}$$

Therefore:

$$[(3I_2)A]_{ij} = 3A_{ij}$$

No pivot argument. No geometry. No column reasoning.

Why "pivot 3" *sounds* convincing but is sloppy

Row-reduction intuition says:

- A pivot in row i picks out row i
- Scaling the pivot scales that row

That intuition is **derived** from the delta identity above.

It is **not** a justification on its own.

Clean logical chain (no interpretation)

1. Identity matrix definition: $(I_2)_{ik} = \delta_{ik}$
2. Scalar multiplication: $(3I_2)_{ik} = 3\delta_{ik}$
3. Matrix multiplication definition
4. Delta identity collapses the sum
5. Result: every entry multiplied by 3

That's it. Nothing else is allowed.

Sources

- Axler, *Linear Algebra Done Right*, Section 3.A
- Strang, *Introduction to Linear Algebra*, Section 2.2

- Wikipedia, "Kronecker delta" — explains the sum collapse
https://en.wikipedia.org/wiki/Kronecker_delta ↗
- Wikipedia, "Matrix multiplication"
https://en.wikipedia.org/wiki/Matrix_multiplication ↗

If you want, I can rewrite the argument **purely in summation symbols**, zero English.



so by the proof you just gave me, thats why we apply the pivot scaled by a constant to each column in A, thus the c is 3, and because it is we do that. Yes or no.

Answer: Yes — but with a strict caveat.

Yes as a conclusion, not yes as a reason.

Precise logical status

- **What is true:**
Left-multiplying by $3I_2$ scales every column of A by 3.
- **Why it is true:**
Because

$$[(3I_2)A]_{ij} = \sum_k 3\delta_{ik}A_{kj} = 3A_{ij}$$

- **What is *not* valid logic:**
"There is a pivot equal to 3, therefore we scale each column."

That statement describes the **effect**. not the **cause**.

Robot-clean logic chain

1. Matrix multiplication is defined entrywise by a sum.

2. $3I_2$ has entries $3\delta_{ik}$.
 3. The delta forces exactly one term of the sum to survive.
 4. The surviving term is multiplied by 3.
 5. Therefore every entry of A is multiplied by 3.
 6. Columns scale as a consequence.
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One-line verdict

Yes, multiplying by $3I_2$ scales each column of A by 3 —

No, the pivot argument is not the justification; it's only an interpretation.

Sources

- Axler, *Linear Algebra Done Right*, §3.A (definition-first proofs)
- Strang, *Introduction to Linear Algebra*, §2.2 (identity action on matrices)
- Wikipedia, "Kronecker delta" — summation collapse

https://en.wikipedia.org/wiki/Kronecker_delta ↗

If you want, I can restate this as a formal "iff" statement to lock the logic completely.



yes iff logic statement please

TL;DR

Left-multiplying by $3I_2$ scales each column of A by 3 **iff** the matrix multiplication is defined using the Kronecker delta identity.

Formal iff statement

Let $A \in \mathbb{R}^{2 \times n}$. Then:

$$(3I_2)A = 3A \iff [(3I_2)A]_{ij} = \sum_{k=1}^2 3\delta_{ik}A_{kj}$$

Proof (both directions)

($\Rightarrow$)

Assume

$$[(3I_2)A]_{ij} = \sum_k 3\delta_{ik}A_{kj}$$

Using the Kronecker delta identity:

$$\sum_k \delta_{ik}A_{kj} = A_{ij}$$

Hence:

$$[(3I_2)A]_{ij} = 3A_{ij}$$

Thus each column (and entry) of A is scaled by 3.

($\Leftarrow$)

Assume left-multiplication by $3I_2$ scales each column of A by 3, i.e.

$$(3I_2)A = 3A$$

Then for every i, j :

$$[(3I_2)A]_{ij} = 3A_{ij}$$

By definition of matrix multiplication, this equality holds **iff**

$$[(3I_2)A]_{ij} = \sum_k (3I_2)_{ik}A_{kj}$$

and since $(3I_2)_{ik} = 3\delta_{ik}$, the summation form follows.

Logical meaning (no intuition)

“pivot scales columns” $\iff$ Kronecker delta collapses the sum

The pivot explanation is *valid iff* it is shorthand for the delta identity.

Sources

- Axler, *Linear Algebra Done Right*, §3.A
- Strang, *Introduction to Linear Algebra*, §2.2
- Wikipedia, “Kronecker delta”
https://en.wikipedia.org/wiki/Kronecker_delta ↗
- Wikipedia, “Identity matrix”
https://en.wikipedia.org/wiki/Identity_matrix ↗

If you want this rewritten in **pure symbolic logic with quantifiers only**, say so.

